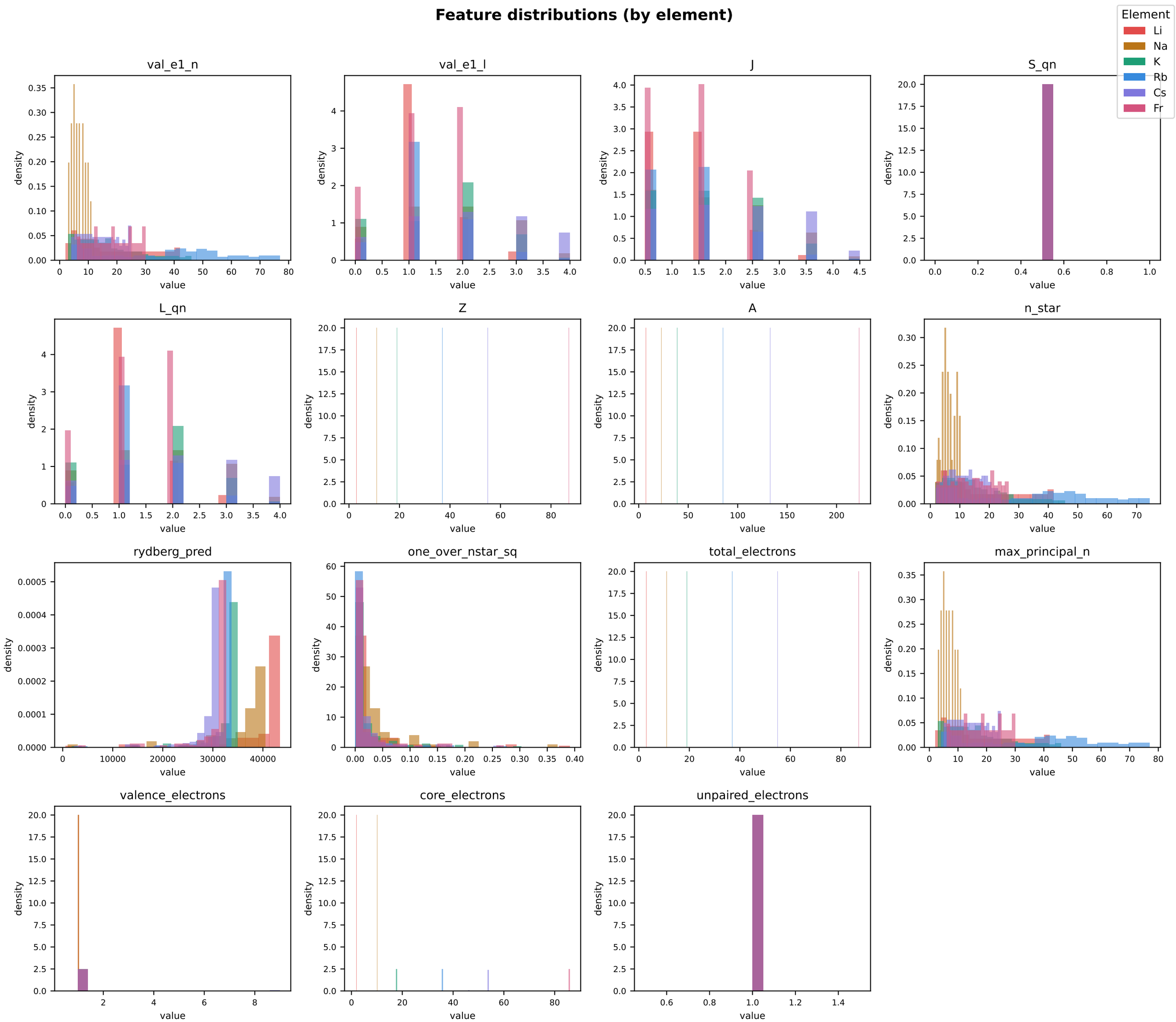
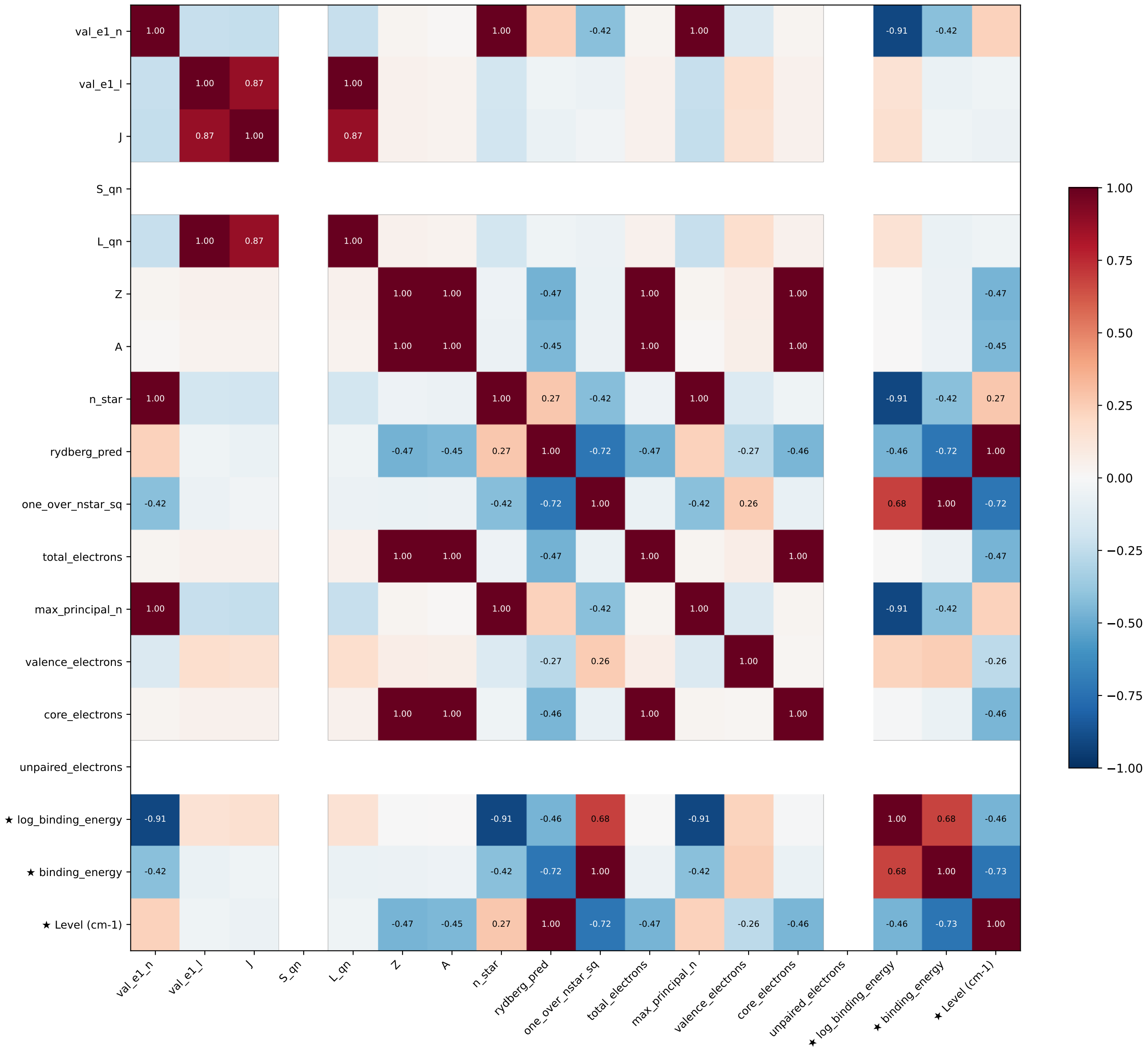
Feature Analysis Report

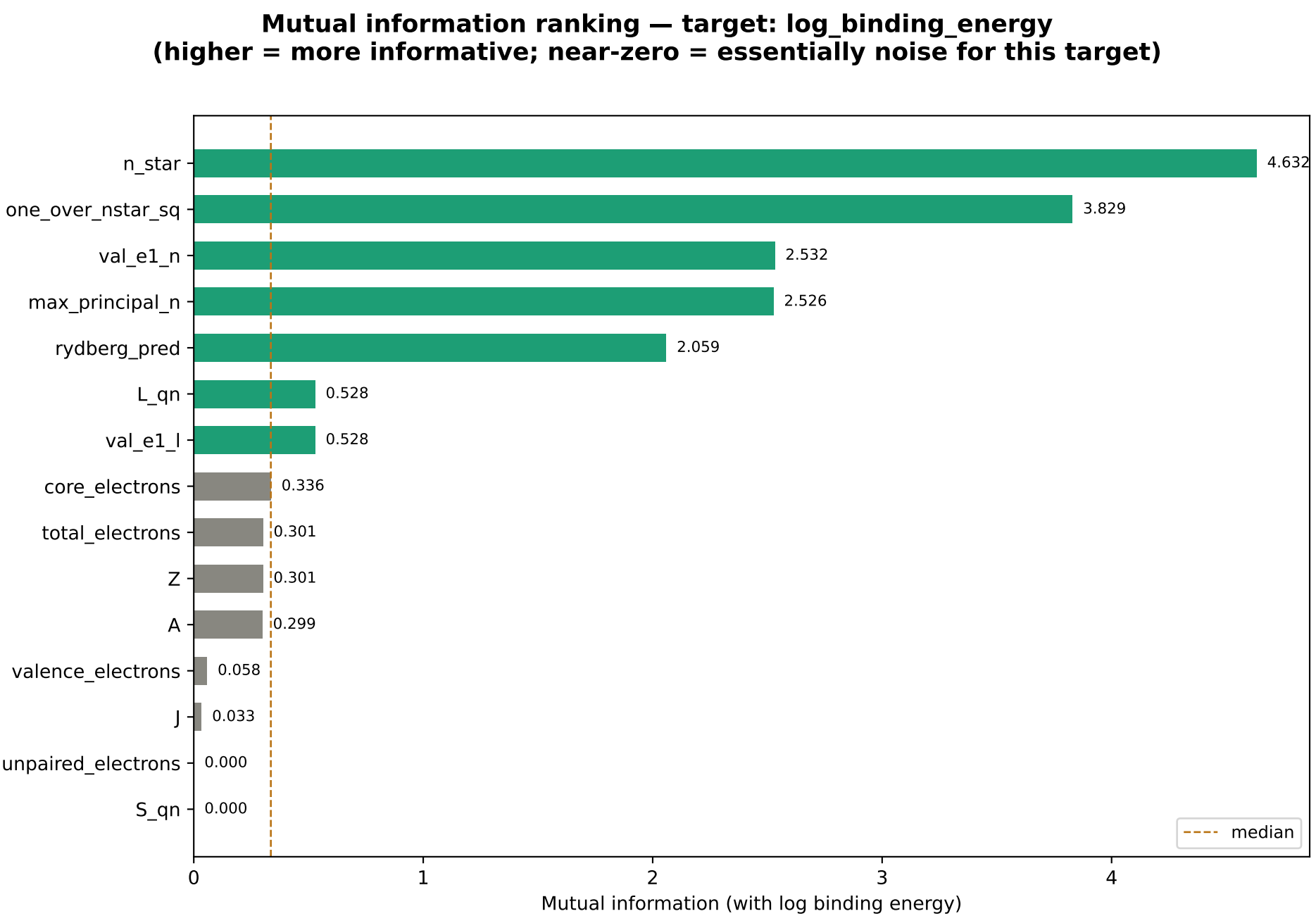
Elements: Li, Na, K, Rb, Cs, Fr · n=844 total samples

Features analysed: val_e1_n, val_e1_l, J, S_qn, L_qn, Z, A, n_star, rydberg_pred, one_over_nstar_sq, total_electrons, max_principal_n, valence_electrons, core_electrons, unpaired_electrons

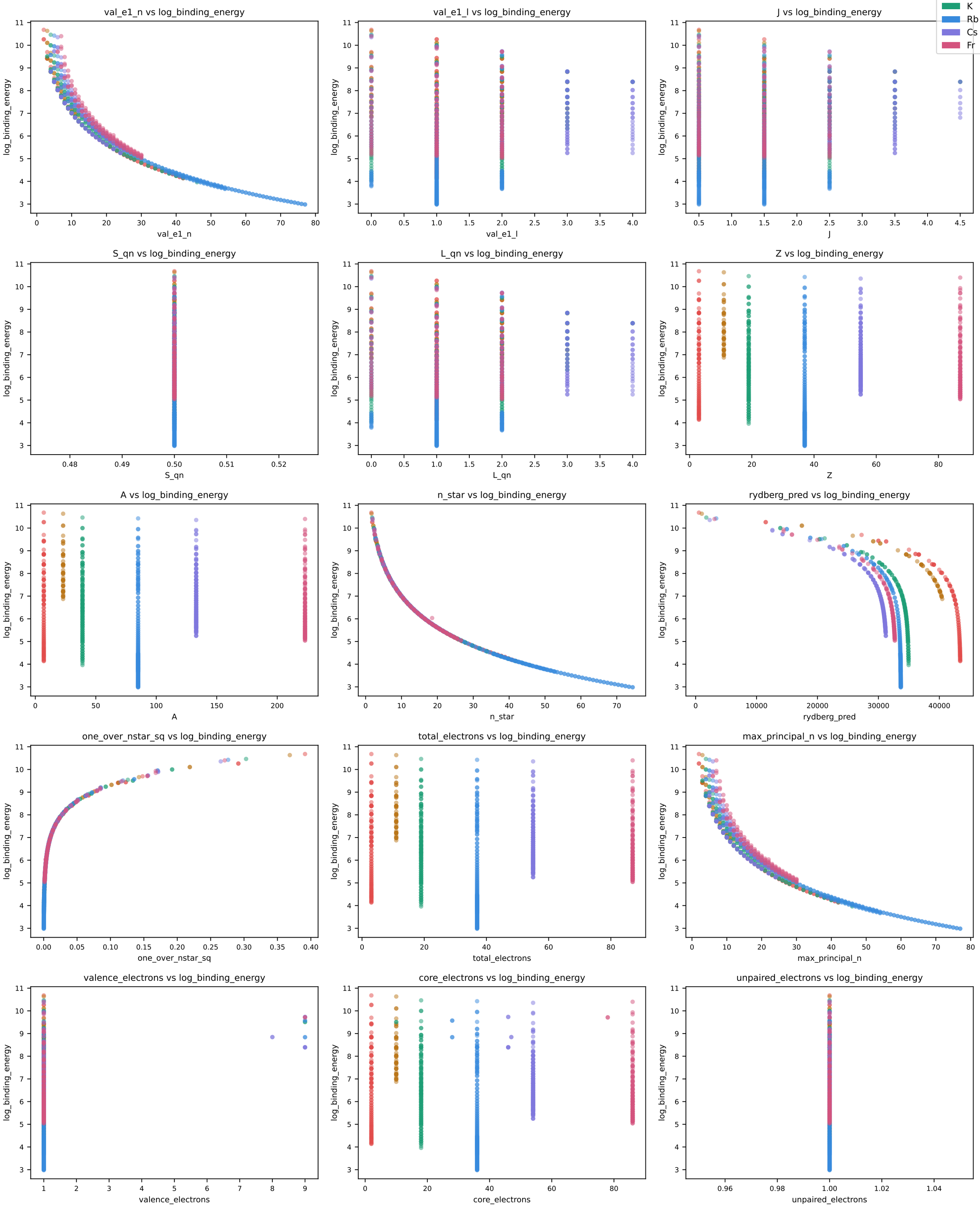

Feature distributions (by element)
Element
Li
Na
K
Rb
Cs
Fr
val_e1_n
val_e1_l
J
S_qn
L_qn
Z
A
n_star
rydberg_pred
one_over_nstar_sq
total_electrons
max_principal_n
valence_electrons
core_electrons
unpaired_electrons
density
value

Feature correlation matrix (★ = target variables)
|r|>0.8 suggests redundancy; look for L_qn≈val_e1_l, S_qn constant

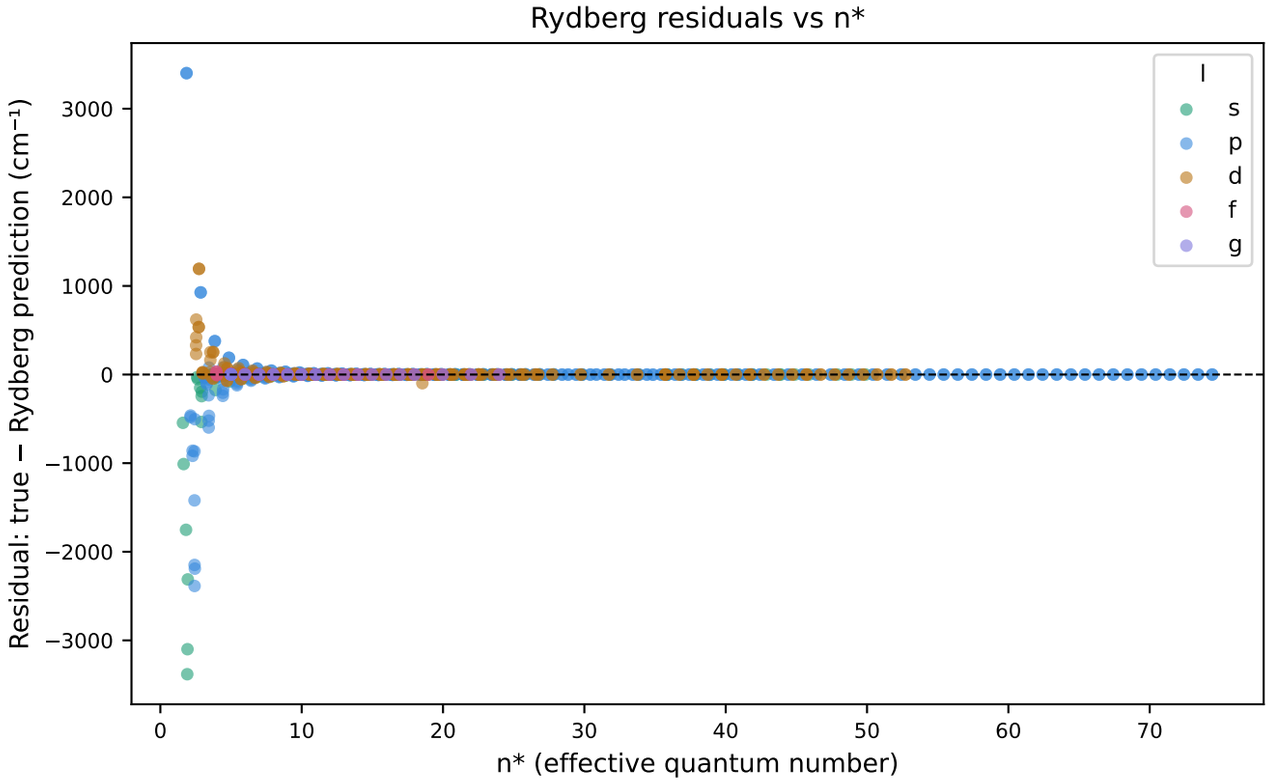


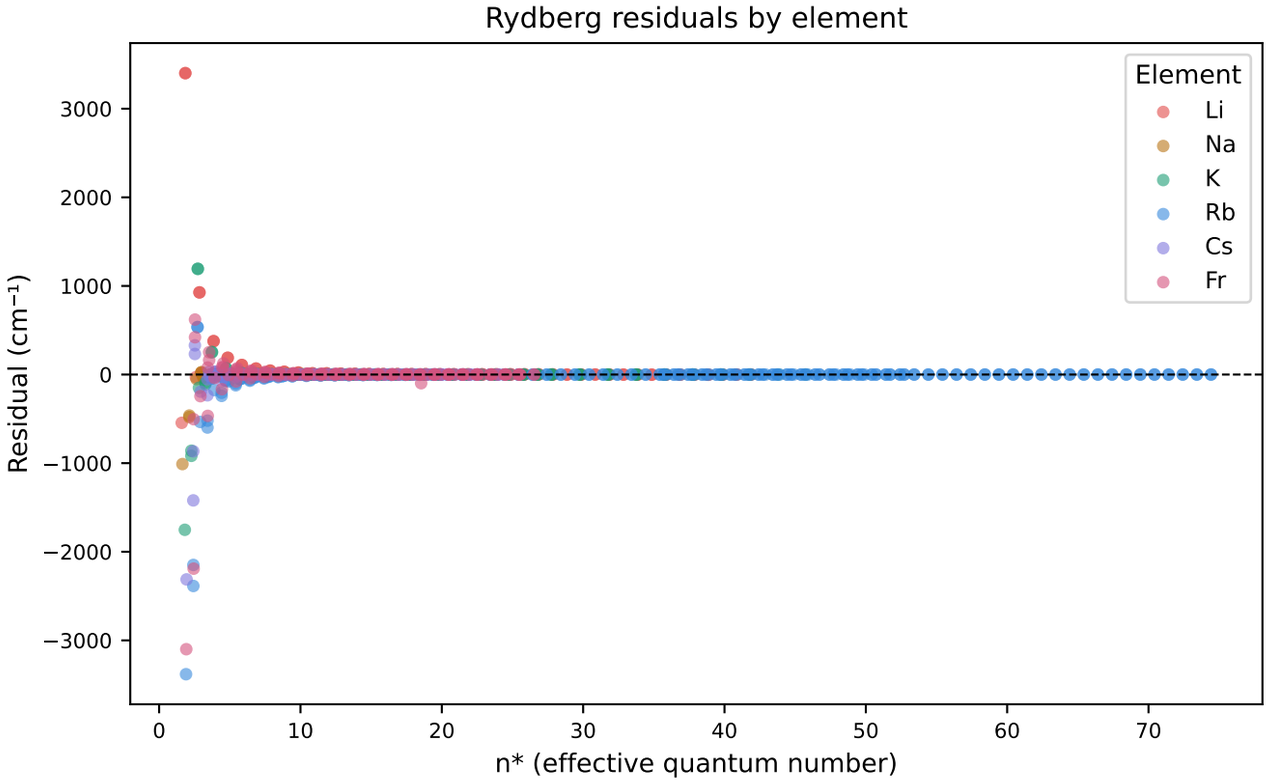

Mutual information ranking — target: log_binding_energy
(higher = more informative; near-zero = essentially noise for this target)
n_star
one_over_nstar_sq
val_e1_n
max_principal_n
rydberg_pred
L_qn
val_e1_l
core_electrons
total_electrons
Z
A
valence_electrons
J
unpaired_electrons
S_qn
4.632
3.829
2.532
2.526
2.059
0.528
0.528
0.336
0.301
0.301
0.299
0.058
0.033
0.000
0.000
median
0
1
2
3
4
Mutual information (with log binding energy)

Feature vs target (log_binding_energy) scatter plots

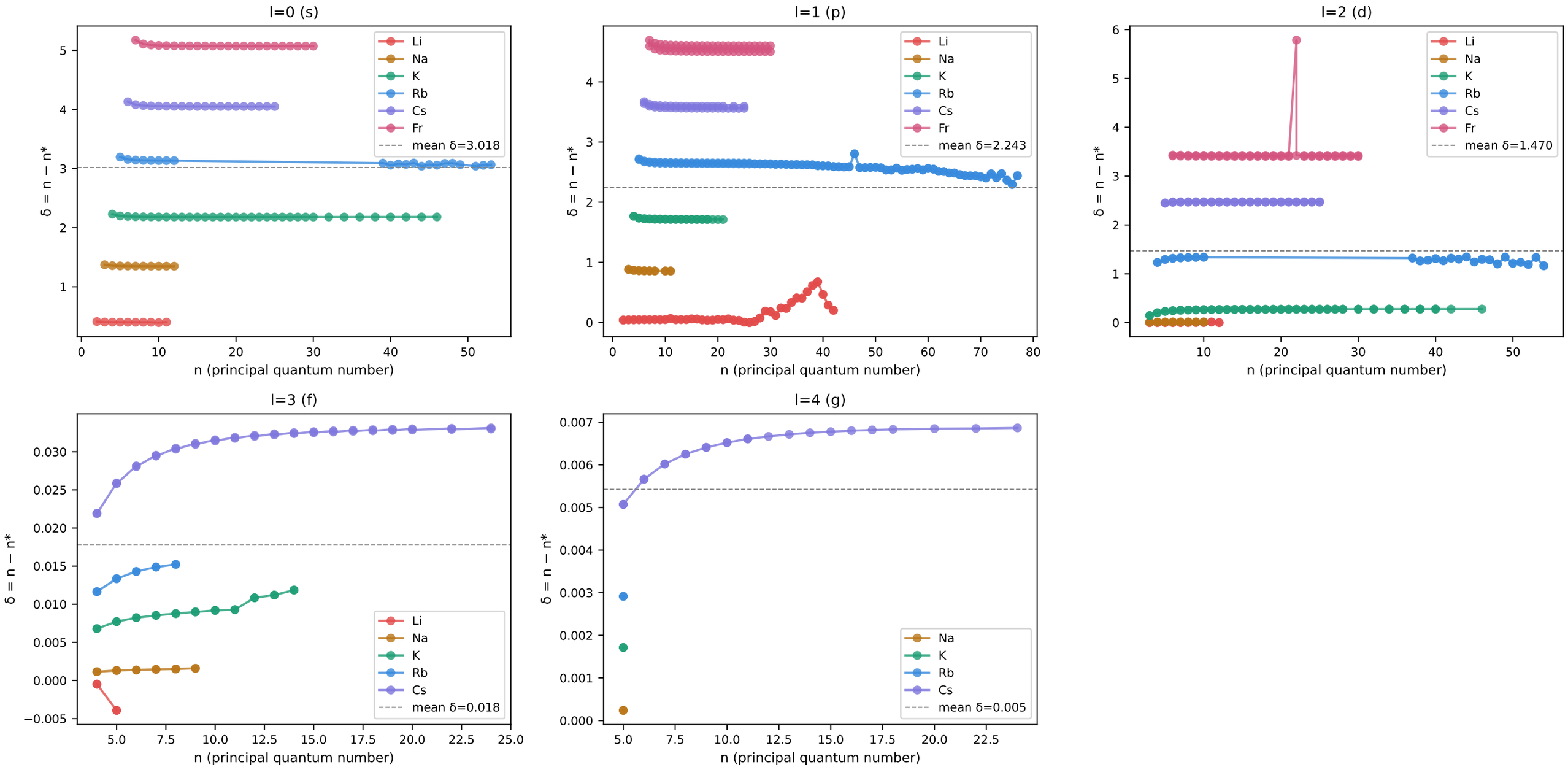


Rydberg residuals (true level − Rydberg prediction)
Large residuals at low n* = extrapolation challenge; ideally residuals should be small and symmetric around 0

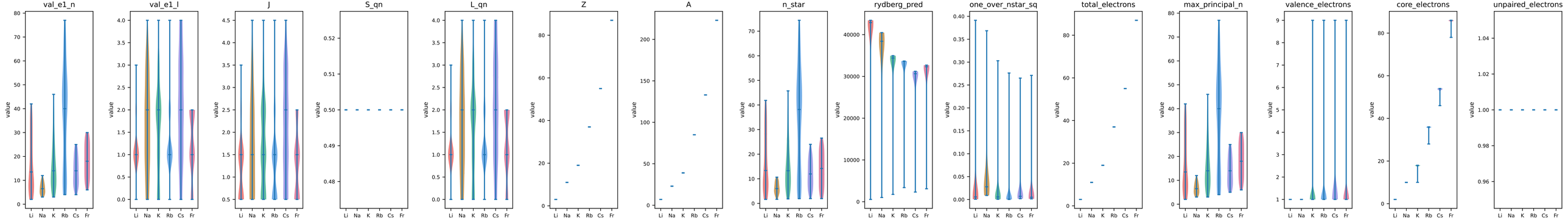
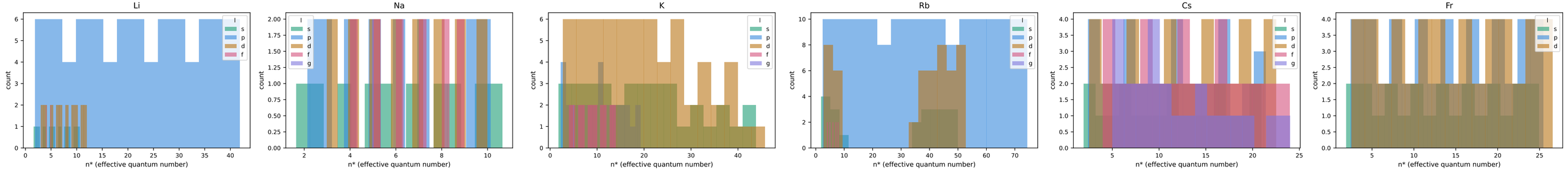




Quantum defect δ = n − n* per angular momentum series
Ideally a flat horizontal line; slope indicates Rydberg-Ritz correction needed



Feature value ranges per element (violin plots)
Mismatched ranges across elements can confuse StandardScaler — look for outliers



n* (effective principal quantum number) distribution per element and series
Gaps at low n* identify states where Rydberg extrapolation is required